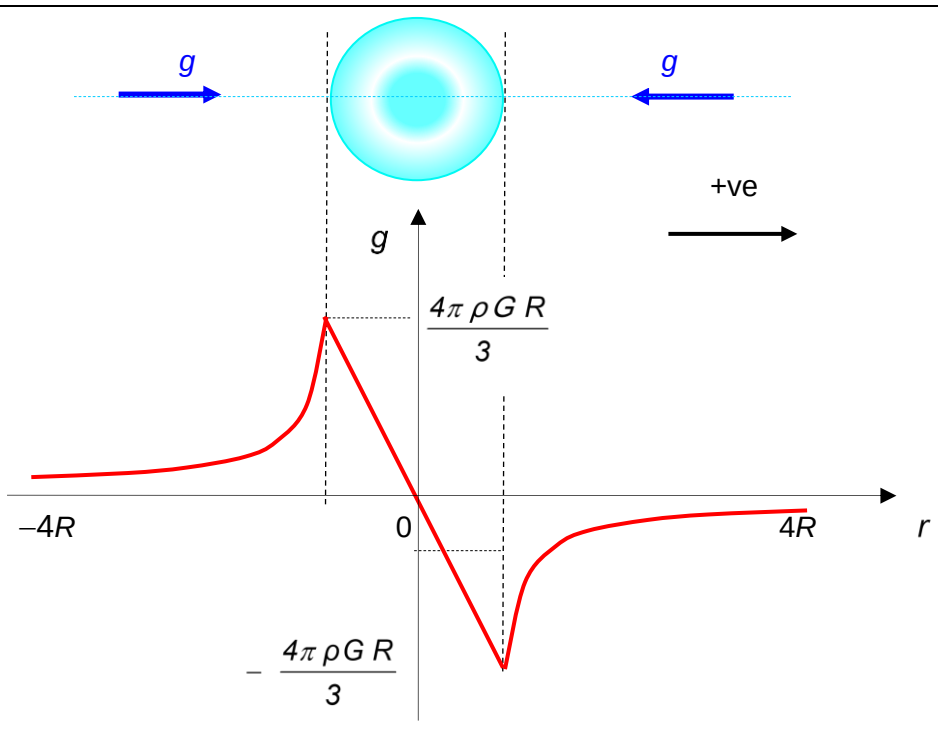


|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| (a) | $g = \frac{F}{m} \quad \text{where } F = \frac{GMm}{r^2}$ $= \frac{GM}{r^2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|     | $\text{as } \rho = \frac{M}{\frac{4}{3}\pi R^3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|     | $g = \frac{4\pi\rho G R^3}{3 r^2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| (b) |  <p>The diagram illustrates the gravitational field strength <math>g</math> as a function of distance <math>r</math> from the center of a uniform sphere of radius <math>R</math> and density <math>\rho</math>. The sphere is shown at the top with two blue arrows labeled <math>g</math> pointing towards it. Below the sphere, a graph plots <math>g</math> on the vertical axis against <math>r</math> on the horizontal axis. The graph shows a red curve that is constant at <math>\frac{4\pi\rho G R}{3}</math> for <math>r &lt; -R</math>, increases to a peak at <math>r = -R</math>, decreases linearly through the origin <math>(0,0)</math> to a trough at <math>r = R</math>, and then increases back to <math>\frac{4\pi\rho G R}{3}</math> for <math>r &gt; R</math>. The region between <math>r = -R</math> and <math>r = R</math> is labeled as a straight line passing through the origin.</p> <p><math>-R &lt; r &lt; R</math>: straight line passing through the origin</p> |

|         |                                                                                                                                                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | $r < -R$ and $r > R$ : inverse square graph<br>+ve and -ve direction of $g$ (according to stated sign convention)<br>$r = R: g = \frac{4\pi \rho G R}{3}$                                                                                                                                                     |
| (c)(i)  | <p>By N2L, <math>ma = mg</math></p> $a = \frac{4\pi \rho G}{3} r \quad \dots\dots(1)$ <p>For SHM, <math>a = \omega^2 x \quad \dots\dots(2)</math></p> <p>Comparing <math>\omega^2 = \frac{4\pi \rho G}{3}</math></p> $\left(\frac{2\pi}{T}\right)^2 = \frac{4\pi \rho G}{3}$ $T = \sqrt{\frac{3\pi}{\rho G}}$ |
| (c)(ii) | <p>Maximum time taken (as assuming released from rest in Singapore) = half a period</p> $\frac{T}{2} = \frac{1}{2} \sqrt{\frac{3\pi}{\rho G}}$ $= \frac{1}{2} \sqrt{\frac{3\pi}{(5.51 \times 10^3) 6.67 \times 10^{-11}}}$ $= 2530 \text{ s (3 s.f.)}$                                                        |
| 5       |                                                                                                                                                                                                                                                                                                               |